# Messages, Not Smoothness

The precision-message graph: every edge a contract, every contract a Gaussian factor, and the global solve as the accountant

Vol-Fitter Technical Notes, No. 14

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**Abstract**

The Vol-Fitter's headline differentiator — filling a mostly-dark universe of smiles from a handful of lit calibrations — has been rebuilt around a new propagation operator, and this note documents the revamped system from the ground up. The original engine regularized: a directed smooth-field prior whose local behaviour *emerges* from global energies, retained here byte-identical and still the shipped default. The new primary operator *routes*: every edge is a contract $z_i \approx \beta\, z_j$ at a stated relation variance, assembled as one Gaussian pairwise factor per relation, and every desk-facing behaviour is a theorem with a golden number locked before the code existed — a receiver inherits its informer's move at full configured amplitude regardless of confidence (fig. 1), competing messages are precision-weighted and averaged, never added (equal opposing signals cancel to 0.00; a $3p$ source outvotes to $-0.50$), a chain multiplies amplitudes while accumulating variance (the mean pays no toll), and a dead informer costs exactly nothing — the defect that rejected the spec's original row-normalized form, which dilutes a lit message to 0.40 against an equal-precision dead neighbour. Amplitude itself splits into a locked maturity *shape* ($\alpha_T = 1$) and a measured *level*: per-class multipliers mechanized through a node-linked innovation anchor whose corroboration behaviour was validated on stored benchmark rows with zero free parameters — calibrated only on the one-source transfer slope (0.391), it predicts the two-source slope 0.563 against a measured 0.561, agreement to 0.3%. The information-form posterior is the accountant: repeated routes from one source are priced jointly ($1.67/p$ against the naive $1.20/p$), disconnected components honestly keep zero innovation and broad bands, and baseline uncertainty enters exactly once per node. A pre-registered six-condition adoption gate and a combined adjudication campaign — absorbing the parked learned-betas sweep — are in flight as this note is written; until that verdict, precision messages ship as an explicit opt-in mode beside the untouched legacy field.

# Contents

## 1 A universe, mostly dark

The desk owns a universe of smiles — one node per (underlying, $T$) — and on any given morning only a few of them are richly quoted. The production spine that answers this is unchanged by the revamp and worth restating in one breath: every node receives a *transported prior* baseline (Note 13's saved prior, moved to today's forward under the spot-dynamics regime of Note 12, through a strict provenance hierarchy); every *lit* node contributes the innovation $d = \text{calibrated} - \text{transported prior}$, a genuine market-versus- prior move fitted in pure-market mode; the graph spreads those innovations to the dark nodes as a posterior increment field $\widehat{z}$ with a marginal uncertainty; and each dark node's absolute handles $h^+ = h^0 + \widehat{z}$ retarget into Note 01's arbitrage-free smile machinery, where residual dark quotes may score the reconstruction but never steer it. The carrier is the compact handle triple $(\sigma_0, s_0, \kappa_0)$ — ATM vol, skew, curvature — solved as three independent fields.

What changed is the middle of the spine: *how* the innovations propagate. The original operator (section 2) is a smooth-field prior — mathematically proper, measurably skilful, and still the shipped default — whose local behaviour is an emergent property of global energies. The revamped primary operator (section 3 onward) is built from explicit relation contracts, because the desk kept asking questions the smooth field could not answer directly: *if I say a 6M move implies twice the move at 3M, will it? If I halve my confidence in that relation, does the predicted move shrink (it must not) or the band widen (it must)? When two sources corroborate, does trust add?*

**Invariant**

1. A dark node stays at its transported prior unless lit data forces a move; a component with no lit path keeps zero innovation, honestly broad bands, and a `no_lit_path` flag — no invented signal, ever.

2. Propagation semantics are *contracts*: amplitude is what the move should be, precision is how much the relation is trusted, and the two never mix. Every contract in sections 3 and 4 is locked by a golden test written before the implementation.

3. One source is one source: however many routes carry it, the global solve prices them jointly — precision never accrues by path counting.

4. Every filled node reports its uncertainty, which widens with graph distance and with informer uncertainty; reported confidence is always the *marginal* posterior precision, never a conditional.

5. The output retargets into the arbitrage-free smile machinery of Note 01; the graph never emits a raw curve.

6. The legacy smooth field remains available and byte-identical at its defaults; mode changes are explicit, and defaults change only on the pre-registered adjudication gate of section 9.3.

**Conventions and the notation ledger.** Node indices $i, j$; the *receiver* of a relation is $i$, its *informer* $j$, and every ordered object reads "$j$ informs $i$". One time symbol $T$ (maturity, years). Innovations $z$ are in the receiving handle's units; ATM-vol units are quoted so that 0.01 is one vol point. Subscripted $\partial$; primes never used.

| | | | |
|---|---|---|---|
| $z_i,\ d_s$ | innovation field; lit observations | $p_{ij},\ \beta_{ij}$ | edge precision; per-handle amplitude |
| $Q_{\text{msg}},\ q_i$ | the factor operator; receiver conditional precision | $\alpha_T,\ \rho$ | amplitude shape exponent; class level |
| $\kappa_i$ | node-linked innovation anchor | $r^d_s,\ p^0_i$ | innovation obs. precision; baseline precision |
| $Q^+,\ \Sigma^+,\ G$ | posterior information; covariance; gain | $g_i$ | cycle gauge potential ($\beta_{ij} = g_i/g_j$) |
| $Q_\Delta,\ L^\beta_{\text{dir}}$ | legacy smooth-field precision; directed residual | $\eta,\ \lambda,\ \nu$ | legacy weights (smoothness, OT, source) |

Table 1: Every symbol in the note. $p$ is always an edge's relation precision; baseline precision is $p^0$, never bare $p$.

## 2 The smooth field, and what it cannot say

The legacy operator deserves an honest summary, because it stays in production and its strengths set the bar. Raw nonnegative trust weights $w_{ij}$ ("$j$ informs $i$") are row-normalized into a stochastic kernel $K$, a stationary mass $\pi$ solves $\pi^\mathsf{T} K = \pi^\mathsf{T}$, and one handle's increment field receives the proper Gaussian prior $z \sim \mathcal{N}(0, Q_\Delta^{-1})$ with

$$Q_\Delta = D_\kappa + \eta\, L^\beta_{\text{dir}} + \lambda\,(A_\rho + \nu I)^{-1}, \qquad L^\beta_{\text{dir}} = (I - K \circ \beta)^\mathsf{T}\, \Pi\, (I - K \circ \beta), \tag{1}$$

a three-member committee: local smallness $D_\kappa$ (which makes the prior proper), directed neighbour prediction $\eta L^\beta_{\text{dir}}$ (PSD for any real $\beta$, being a Gram matrix in the $\Pi$-inner product), and an optional unbalanced-optimal-transport tangent term (shipped off, $\lambda = 0$). Conditioning on the lit innovations is done in covariance form — only the observed columns and one $n_{\text{obs}} \times n_{\text{obs}}$ solve — and the reported confidence is the reciprocal *marginal* posterior variance $1/K^+_{ii}$, never the precision-matrix diagonal, which overstates certainty exactly at the dark nodes whose neighbours are themselves uncertain. The same covariance algebra yields the closed-form observation-selection gain (which dark quote removes the most exposure-weighted variance) and the exact per-source attribution. All of this is retained, byte-identical, as `smooth_field` mode — and its measured skill is real (section 9).

But four structural facts stop the smooth field from expressing the desk's propagation contracts, and each is a one-line observation about (1):

1. **Row normalization discards absolute information.** Incoming weights $(p, p)$ and $(100p, 100p)$ normalize to the same row — the *relative* average survives, the fact that one receiver hears a hundred times more total information does not. "Independent precisions add" is not expressible.

2. **Strength and confidence share one dial.** Raising $\eta$ makes neighbour prediction more influential *and* the whole field stiffer; "full mean transmission with deliberately wide uncertainty" has no configuration.

3. **The anchor taxes valid messages.** A receiver with anchor $\kappa$ and incoming precision $p$ transfers only $p/(\kappa + p)$ of a correct message; removing the tax by raising $p$ simultaneously (and wrongly) raises confidence.

4. $\pi$ **is not a meaning.** The stationary mass is the right geometry for a smoothness energy, but it answers no desk question, and on one-way topologies it can silently zero out whole node classes — the transient-name case file of section 9 is exactly this failure surfacing in a harness.

These are not tuning problems; they are the operator asking a different question. The smooth field asks "what small, neighbour-consistent, transportable field explains the lit moves?" The desk was asking "what did each relation *say*, how much do I trust it, and what do the messages add up to?" The revamp answers the second question directly.

## 3 Edges as contracts

**Definition 1** (Message edge)**.** A directed relation from informer $j$ to receiver $i$ consists of a conditional relation precision $p_{ij} > 0$, a per-handle amplitude triple $\beta_{ij}$, and a relation class (calendar, broad index, sector ETF, sector peer, custom). Its meaning is the Gaussian statement

$$z_i = \beta_{ij}\, z_j + \epsilon_{ij}, \qquad \epsilon_{ij} \sim \mathcal{N}(0,\, 1/p_{ij}) :$$

*amplitude says what the receiver's move should be; precision says how much the relation is trusted. They never mix.*

Each edge contributes one rank-one Gaussian factor, and the propagation operator is their sum:

$$\boxed{Q_{\text{msg}} = \sum_{(j\to i)} p_{ij}\, u_{ij} u_{ij}^{\mathsf{T}}, \qquad u_{ij} = e_i - \beta_{ij} e_j, \qquad q_i = \sum_j p_{ij}.} \tag{2}$$

Positive semidefiniteness is immediate for arbitrary real betas (a sum of PSD rank-one terms), and the assembly is sparse by construction — each factor touches two nodes.

**Proposition 1** (Receiver conditional)**.** *Minimizing the factor energy over $z_i$ with the informers held fixed gives*

$$z_i \mid \{z_j\} \sim \mathcal{N}\left(\frac{\sum_j p_{ij}\beta_{ij} z_j}{q_i},\ \frac{1}{q_i}\right).$$

*Incoming messages are precision-weighted and* averaged*; independent incoming conditional precisions* add.

*Proof.* The terms of (2) containing $z_i$ are $\sum_j p_{ij}(z_i - \beta_{ij} z_j)^2$; completing the square in $z_i$ gives the stated mean and variance. □

This is the entire desk semantics in one line, and fig. 1 shows its sharpest consequence through the production solve: on the three-expiry ladder (3M/6M/1Y, 6M lit +1 vol point, $\alpha_T = 1$) the dark receivers land at exactly $\widehat{z}_{3M} = +2.00$ and $\widehat{z}_{1Y} = +0.50$ for *every* edge precision across three decades, while the posterior standard deviation scales as $p^{-1/2}$. Amplitude is not a function of confidence — the first thing the smooth field could not promise, delivered as an identity.

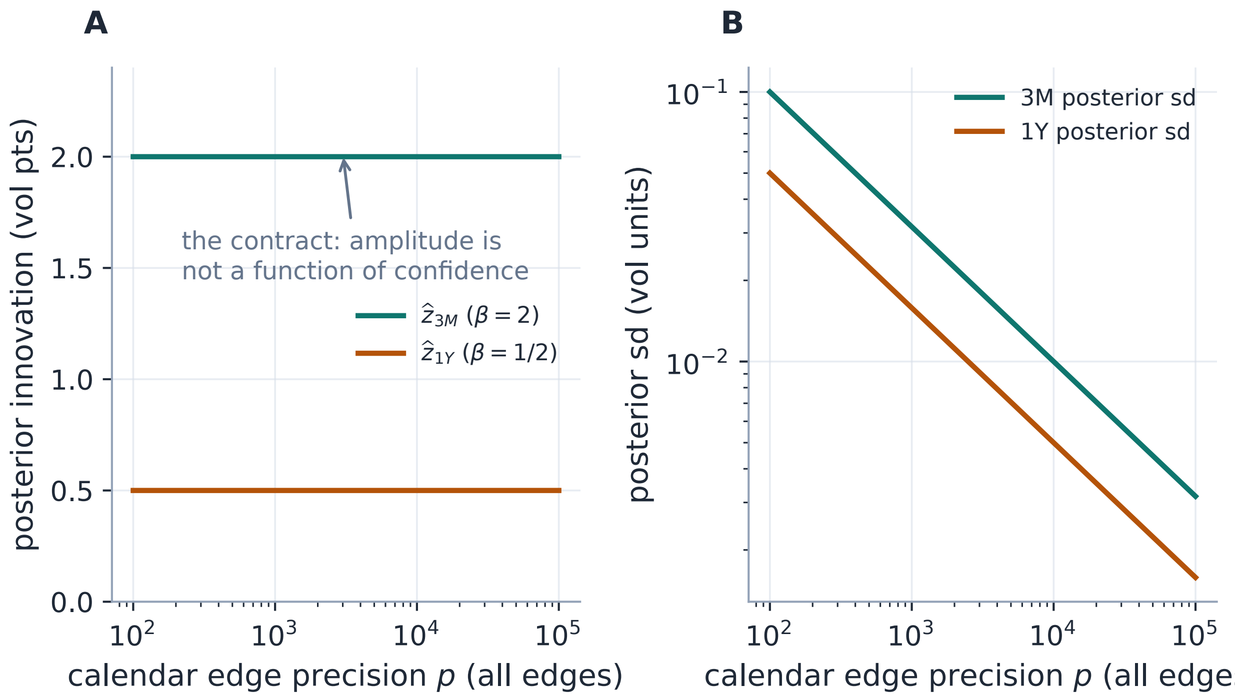


Figure 1: The contract, run through the production operator and posterior (three-expiry ladder, 6M lit +1 vol point). A: the dark receivers' posterior means sit at the configured amplitudes +2 and +0.5 regardless of edge precision. B: what the precision *does* control — the posterior standard deviation, falling as $p^{-1/2}$. Confidence and amplitude are separate dials because they are separate fields of the edge.

### 3.1 Why pairwise factors: the operator that diluted its best source

The specification's first draft boxed a different assembly: row-normalize the incoming precisions, $K^p_{ij} = p_{ij}/q_i$, and charge each receiver $q_i\,(z_i - \sum_j K^p_{ij}\beta_{ij}z_j)^2$. Both forms produce the identical receiver conditional of proposition 1, so every local golden contract is blind to the choice. They differ in the *joint* distribution, and the difference has teeth.

**Case file: the dead informer**

**Setup.** A receiver hears two informers at equal configured precision: one lit, one *dead* — a dark dead-end with no lit path and no other relations.
**Failure mode (row form).** The averaging weights are fixed by *configured* precision even when an informer carries no information: the row residual can be zeroed for any $z_i$ by moving the free informer, so the posterior is improper, and under a small regularizing anchor the receiver transfers only 0.40 of the lit message at equal precisions — decaying toward zero as the dead informer's configured precision grows (fig. 5A). Configuring trust in a silent neighbour *destroys* a live signal.
**The pairwise form.** Marginalizing an unconstrained informer removes its factor exactly — the Gaussian integral over a free $z_j$ of $p\,(z_i - \beta z_j)^2$ is flat in $z_i$ — so the lit message passes at full amplitude (production: transfer 1.00 across four decades of dead-informer precision), the dead node simply rides along with a broad marginal, and no improper direction survives the

component solve.

**Two more reasons, one convention.** A bidirectional reciprocal pair collapses to *one* factor, so auto-generated calendar relations cannot double-count precision; and the factor list *is* the sparse assembly. The convention that makes one-factor-per-relation well defined is the canonical orientation: $p\,(z_i - \beta z_j)^2$ and $p'\,(z_j - z_i/\beta)^2$ are the same factor iff $p_{\text{rev}} = p_{\text{fwd}}/\beta^2$, so auto calendar relations are quoted with the *shorter* maturity as receiver (relation noise in short-dated vol units, where desk intuition lives), the implied reverse amplitude is $1/\beta$, and the receiver diagnostic $q_i$ maps every incident factor into $i$'s units by that identity. Explicit user edges stay directed as entered — entering both directions deliberately creates two relations, and the cycle diagnostics of section 7 apply. *The joint form of an operator is a modelling decision even when every local conditional agrees.*

# 4 Amplitude: shape locked, level measured

## 4.1 The maturity shape

For a calendar relation the amplitude is the locked maturity shape

$$\beta_{i\leftarrow j} = \Big(\frac{T_j}{T_i}\Big)^{\alpha_T}, \qquad \alpha_T = 1 \text{ (default, per-handle configurable).} \tag{3}$$

$\alpha_T = 1$ is constant total-variance injection — a +1 vol-point move at 6M reads as +2 at 3M and +0.5 at 1Y (fig. 1) — and reciprocal by construction: $\beta_{i\leftarrow j}\beta_{j\leftarrow i} = 1$, so a calendar ladder cannot claim amplification both ways. The exponent is per-handle (ATM level, skew and curvature need not share a maturity scaling; all default 1), and the Phase-0 study found the shape *weakly identified* at the day horizon ($R^2$ 0.180 vs 0.181 across $\alpha_T \in \{0, 1\}$ once the level refits) — so 1.0 is retained for its semantics, and the adjudication campaign sweeps it.

## 4.2 The level is an empirical quantity

Full-force transmission ($\widehat{z} = \beta z$ exactly) is the correct semantics *conditional on a relation the desk asserts*. It is not what the tape does on average: on ∼11735 stored adjacent-pair observations the realized day-over-day calendar transfer under the $\alpha_T = 1$ shape is 0.23 per unit of predicted move, and the one-source index→name transfer is 0.391. A default that transmits at full force would fail the pre-registered RMS gate on its first day. So amplitude splits: the *shape* stays locked, and the *level* is a per-relation-class multiplier $\rho \in (0, 1]$ with two presets — `desk` ($\rho = 1$, full force, the belief mode) and `learned` (the measured levels above; note they are shape-dependent — 0.34 was the $\sqrt{T}$-shape calendar multiplier, 0.23 is the $\alpha_T = 1$ value).

## 4.3 The two wrong mechanizations, and the right one

How should $\rho$ enter? Two natural answers are provably wrong. Scaling the beta ($\beta \to \rho\beta$) shrinks the forward conditional but *amplifies* the reverse one by $1/\rho$ — the reciprocal identity turns attenuation into amplification. Emitting two directed shrunk factors double-counts the relation and composes to $2\rho/(1 + \rho^2) \neq \rho$. The mathematically consistent mechanization of regression attenuation in both directions of one joint Gaussian is a *local innovation anchor*:

$$\kappa_i = p_{\text{primary}} \frac{1 - \rho_{\text{class}}}{\rho_{\text{class}}}, \tag{4}$$

with $p_{\text{primary}}$ the largest incident relation precision in node $i$'s units and $\rho_{\text{class}}$ that primary relation's class multiplier — *fixed at build time, never rescaled as further edges arrive.* The desk preset $\rho = 1$ gives $\kappa = 0$ exactly, recovering the unamended contracts above.

**Proposition 2** (Corroboration under the node-linked anchor)**.** *With $k$ equal agreeing clamped sources at precision $p$, beta one, and the fixed anchor* (4)*, the receiver's transfer per unit message is*

$$\frac{kp}{\kappa + kp} = \frac{k\rho}{1 - \rho + k\rho} :$$

*one source transfers exactly $\rho$; two lift it to $2\rho/(1+\rho)$; independent corroboration keeps raising the effective transfer toward one.*

*Proof.* By proposition 1 the $k$ agreeing messages average to the common value at conditional precision $kp$; the anchor adds a zero-innovation pseudo-message at precision $\kappa$, so the posterior mean shrinks by $kp/(\kappa + kp)$; substitute (4). $\square$

This is where the redesign earns the word *measured*. The competing mechanization — an edge-linked anchor $\kappa_i = \sum_j p_{ij}(1-\rho)/\rho$, under which the transfer is constant at $\rho$ regardless of source count — makes a different falsifiable prediction, and the stored benchmark rows adjudicate: across 1007 name-days carrying same-sector peers, the one-source index transfer slope is 0.391 and the measured two-source (index + peer-average) slope is 0.561 — a corroboration uplift of +43% against a pre-registered 15% bar. Calibrating the fixed-$\kappa$ model on the *single-source slope alone* predicts 0.563 for two sources: agreement to 0.3%, with zero free parameters (fig. 2). The node-linked anchor is not a preference; it is the mechanization the data chose.

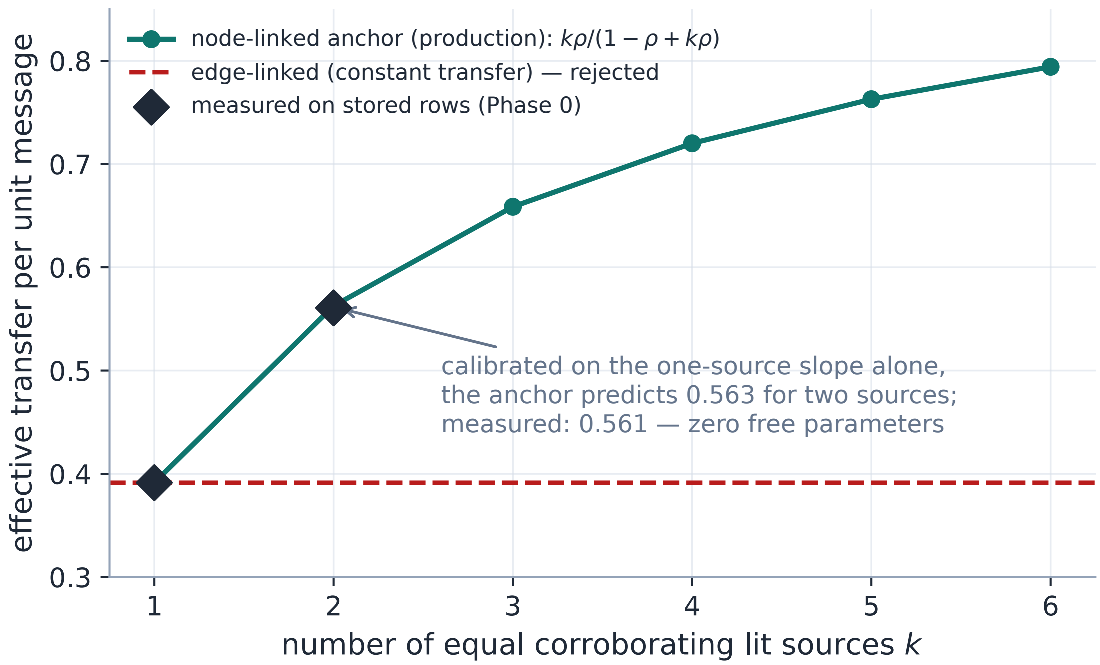


Figure 2: Amplitude level as a measured object (production anchor and posterior; measured points from the stored Phase-0 study artifact). The node-linked fixed anchor produces the corroboration curve $k\rho/(1-\rho+k\rho)$; calibrated only on the one-source slope (0.391), it predicts the measured two-source transfer to 0.3% with no free parameters. The edge-linked alternative — constant transfer regardless of source count (dashed) — is rejected by the same measurement (+43% uplift against a 15% bar).

**Exercise 1.** Derive both rejections of section 4: (i) show that replacing $\beta$ by $\rho\beta$ in one factor turns the implied reverse-direction amplitude into $1/(\rho\beta)$; (ii) show that two directed factors $p(z_i - \rho\beta z_j)^2 + p'(z_j - \rho z_i/\beta)^2$ with matched units produce a one-source transfer of $2\rho/(1+\rho^2)$, not $\rho$. Then verify from (4) that $\rho = 1$ gives $\kappa = 0$ exactly — the desk preset is the contract semantics, not a limit.

## 5 Confidence: the precision families

Amplitude answers "how large"; precision answers "how reliable", and the defaults are empirical. The calendar family is

$$p_{ij}^{\text{cal}} = \frac{p_0}{\epsilon_T + \sqrt{|T_i - T_j|}}, \qquad p_0 \approx 1690 \text{ vol}^{-2}, \quad \epsilon_T \approx 0.97\sqrt{\text{years}}, \tag{5}$$

with the seeds fitted on the same 11735 stored adjacent-pair residuals (fig. 3B): about 2.7 vol points of relation noise at a one-month gap, and — the honest surprise — *nearly gap-flat* at the day horizon: $\epsilon_T$ dominates, the family degrades gracefully toward constant precision, and whether the decay term earns its keep at all is one of the campaign's ablations (`constant` and log-distance families ship beside it). Cross-class seeds from the same study: index→name $p \approx 1.3 \times 10^4$, sector peer $p \approx 0.9 \times 10^4$ (vol units) — with the recorded caveat that both are measured on ticker-day *median* innovations and are therefore upper bounds on per-edge precision.

Two unit disciplines complete the picture. **Per-handle units**: edge precision is quoted in ATM-vol units; the skew and curvature fields scale it by $(s_\sigma/s_h)^2$ with the production per-handle move scales $(0.03, 0.05, 0.5)$ — a units choice, not a semantics choice, since the precision-weighted *average* is invariant to a global rescale of all incoming precisions. **Variance is the readable coordinate**: an edge carries $z_i = \beta z_j + \epsilon$, Var $\epsilon = 1/p$, so along a chain amplitudes multiply while variances accumulate,

$$E[z_C] = \beta_2\beta_1\, E[z_A], \qquad \mathrm{Var}(z_C) = (\beta_2\beta_1)^2\,\mathrm{Var}(z_A) + \frac{\beta_2^2}{p_1} + \frac{1}{p_2}, \tag{6}$$

and *no mean haircut is ever applied for distance* (fig. 4): the mean crosses six hops undamped (production: $\widehat{z} = 1.000$ at hop six) while the band pays the toll, the production marginals matching the accumulation identity to 0.0000 exactly.

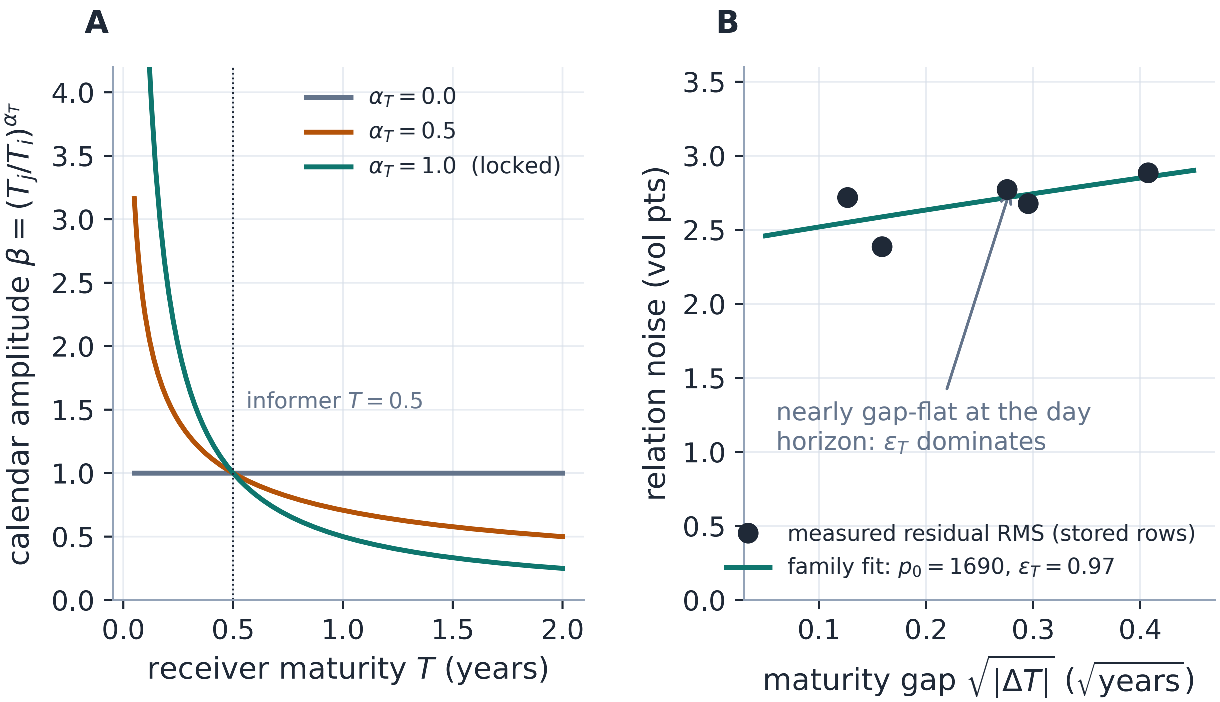


Figure 3: The calendar relation, split into its two fields. A: the amplitude shape (3) — what a 6M move implies elsewhere on the ladder, per exponent; $\alpha_T = 1$ (locked) is constant total-variance injection. B: the confidence family (5) against the stored residuals it was fitted on — relation noise is nearly gap-flat at the day horizon, so $\epsilon_T$ dominates and the decay term's keep is an open ablation.

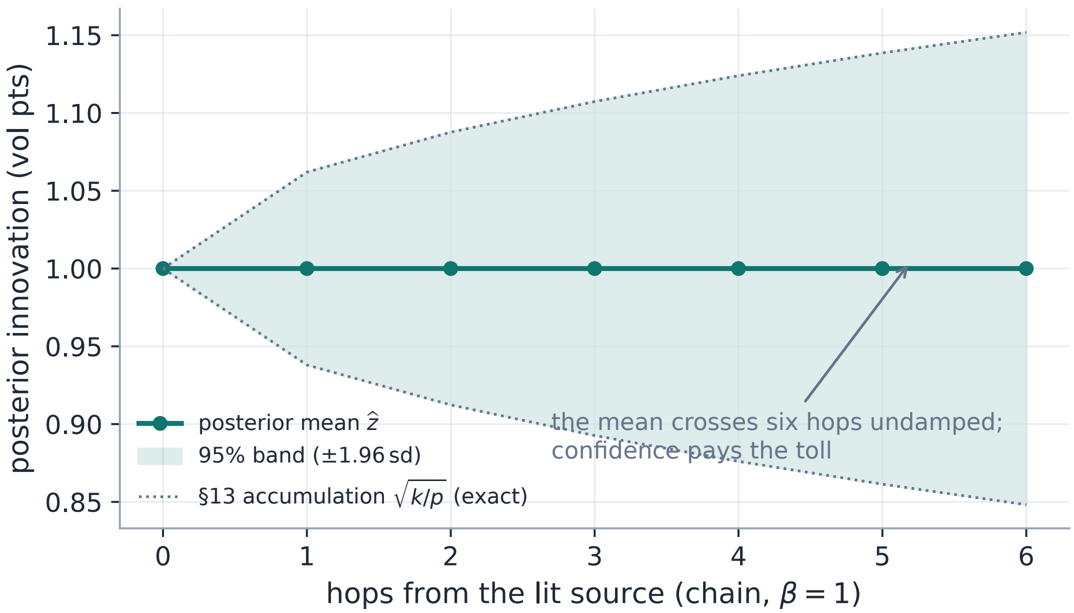


Figure 4: Multi-hop propagation (production solve, six-hop chain, $\beta = 1$): the posterior mean crosses every hop undamped while the marginal band accumulates exactly the per-edge relation variances of (6) (dotted: the closed-form accumulation). Distance costs confidence, never amplitude — the smooth field's coupled dial, uncoupled.

## 6 The accountant: the information-form posterior

The factors, the anchors and the lit observations assemble in information form, solved per connected component of the factor support:

$$\boxed{Q^+ = Q_{\text{msg}} + D_\kappa + H^\top R_d H, \qquad b^+ = H^\top R_d\, d, \qquad \hat{z} = (Q^+)^{-1} b^+, \quad \Sigma^+ = (Q^+)^{-1},} \tag{7}$$

with three honesty rules that are as load-bearing as the algebra.

**No lit path, no invented signal.** Components are built on $\beta \neq 0$ edges (a zero-beta factor couples only its receiver — the reachability guard that stops an information-free informer from destabilizing an observed component); a component with no lit observation is never solved into propriety — it keeps $\hat{z} = 0$, the transported prior, a `no_lit_path` flag, and an explicitly broad variance (one typical handle magnitude at the production layer). Every observed component is verified positive definite by Cholesky before inversion.

**Innovations are only as precise as their ingredients.** A lit innovation is a difference of two estimates, so its observation precision combines harmonically,

$$r_s^d = \Big(\frac{1}{r_s^{\text{cal}}} + \frac{1}{p_s^0}\Big)^{-1}, \tag{8}$$

and the *placement rule* holds baseline uncertainty to exactly one appearance per node: folded into $r_s^d$ for a lit source, added once to the reconstruction band for a dark node — golden-locked so no node is widened twice.

**Attribution is exact.** The gain matrix $G = \Sigma^+ H^\top R_d$ decomposes every posterior shift over the observed lit sources, $\widehat{z}_i = \sum_s G_{is} d_s$, contributions summing to the shift by construction — per independent *source*, never per path, because path contributions are correlated and non-unique.

The accountant's signature behaviours run through the production solve in figs. 5 and 6. Two routes carrying one source are priced jointly: on the triangle fixture (source observed at finite precision $p$, one direct and one two-leg route) the global posterior variance is $1.67/p$ where naive per-message accounting — the eq. (6) effective precisions added as if independent — claims $1.20/p$, overstating precision by 39%. Competing messages vote: equal opposing signals cancel to 0.00 at *doubled* conditional precision $2p$ (disagreement is not silence); a $3p$ source outvotes to $-0.50$; and under $\alpha_T = 1$ betas the same $\mp 1$ raw signals land at $+0.75$ — not a defect but the contract: signals are mapped into receiver units *before* the vote, and equal-absolute-vol cancellation is a $\beta = 1$ statement.

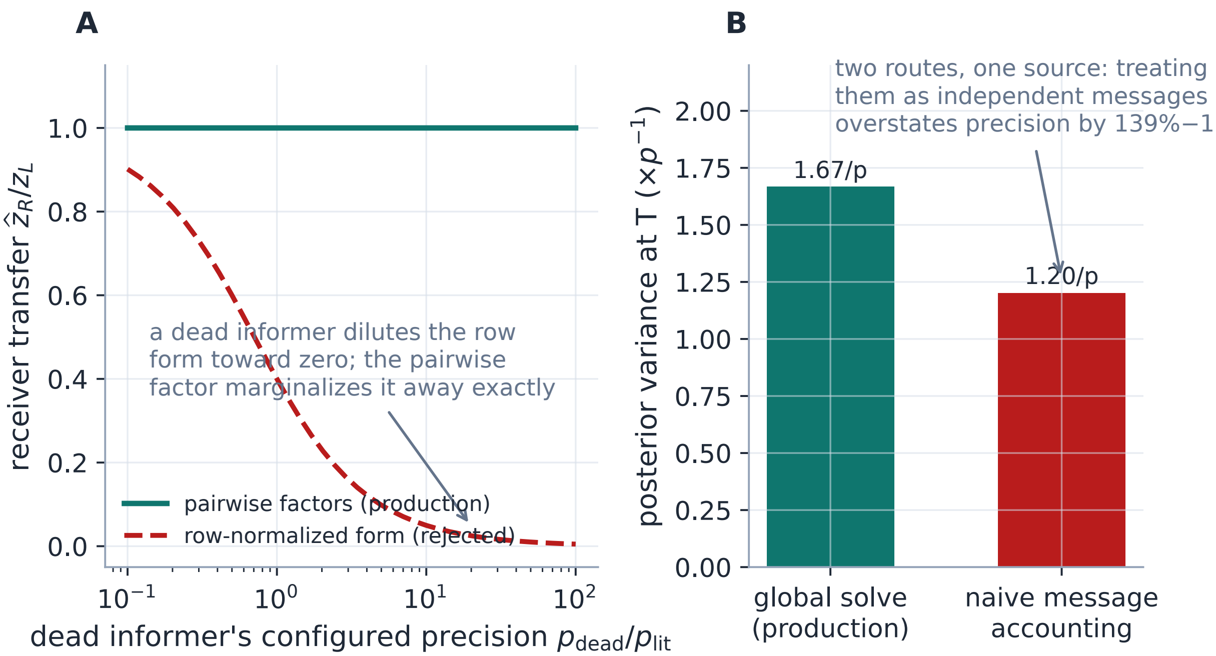


Figure 5: The accountant at work (production solves). A: the dead-informer case file — the rejected row-normalized operator dilutes a lit message toward zero as a silent neighbour's *configured* precision grows; the pairwise factor marginalizes the dead informer away exactly. B: repeated routes from one source — the global posterior variance $(1.67/p)$ against naive independent-message accounting $(1.20/p)$: path counting would fabricate a 39% precision overstatement.

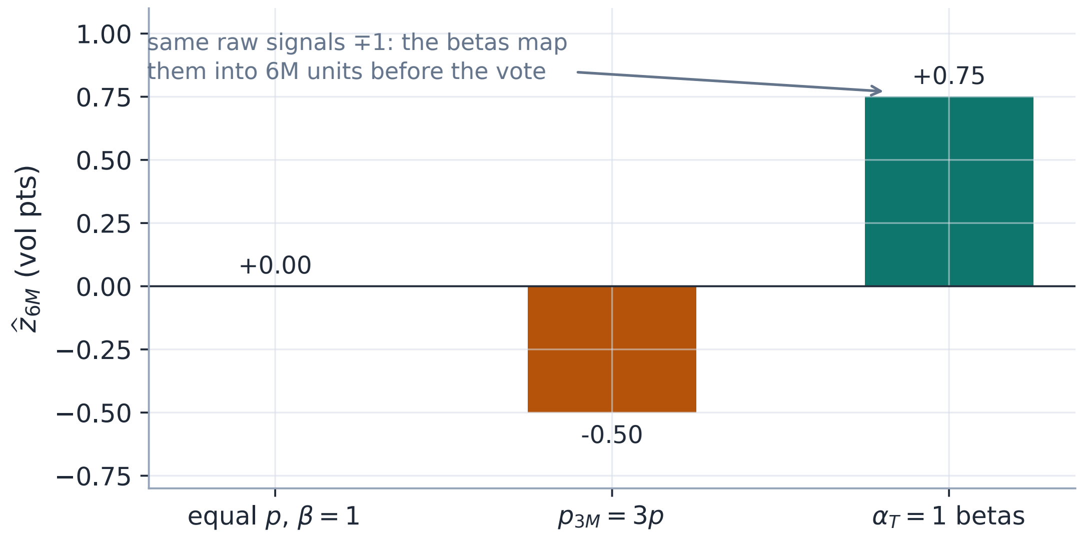


Figure 6: Competing messages vote (production solves; 6M dark, 3M and 1Y lit at $\mp 1$ vol point). Equal precision and $\beta = 1$: exact cancellation at doubled conditional precision. A $3p$ source outvotes. Under the locked $\alpha_T = 1$ betas the same raw signals map to $-0.5$ and $+2$ in 6M units before averaging — the receiver hears receiver-unit predictions, not raw levels.

**Exercise 2.** Reproduce panel B of fig. 5 by hand: with the source observed at precision $p$ and all three relation precisions $p$, assemble the $3 \times 3$ information matrix of (7), invert, and show the target's marginal variance is exactly $5/(3p)$, while the eq. (6) effective-message precisions ($p/2$ direct, $p/3$ through the middle node) sum to $5p/6$, i.e. variance $6/(5p)$. The naive answer is wrong because both routes share the source's own uncertainty — the covariance the global solve refuses to forget.

**Exercise 3.** Three distinct quantities appear at every receiver: edge precision $p_{ij}$, conditional incoming precision $q_i = \sum_j p_{ij}$, and marginal posterior precision $1/\Sigma^+_{ii}$. Construct a two-node example where $q_i$ is large but the marginal precision is small (an uncertain informer), and one where the marginal exceeds $q_i$ (the receiver's own observation). Only in the idealized clamped-independent-informer case do the last two coincide — which is why the wire reports both, and the UI must never present $q_i$ as the final confidence.

## 7 Diagnostics and the gauge

Directedness in a Gaussian is a statement about the *prediction relation*, not about inference: the posterior precision is symmetric, so observing a receiver legitimately updates an uncertain informer. Where strictly one-way conditioning is required, the graph must be a DAG or the source clamped — a UI arrow cannot create one-way Bayes. What the solver *can* police is internal consistency of the amplitudes:

**Proposition 3** (Cycle consistency is a gauge condition)**.** *A positive beta structure is cycle-consistent — every directed cycle's beta product equals one — iff there exist node potentials $g_i > 0$ with $\beta_{ij} = g_i/g_j$.*

*Proof.* If $\beta_{ij} = g_i/g_j$, every cycle product telescopes to one. Conversely, fix a spanning tree per component, set $\phi_i = \log g_i$ by propagating $\log \beta$ along tree edges from an arbitrary root; a non-tree edge with $\beta_{ij} \neq e^{\phi_i - \phi_j}$ closes a cycle whose product is $\beta_{ij} e^{-(\phi_i - \phi_j)} \neq 1$. □

Production runs exactly this proof as an algorithm: a union-find sweep with logarithmic offsets assigns the potentials in linear time and flags every closing edge whose implied cycle product strays from one (the reference cross-check of appendix D plants a consistent triangle — zero flags — and an inconsistent one, flagged at product 0.833). Auto-generated calendar ladders are gauge-consistent by construction ($g_i = T_i^{-\alpha_T}$); the diagnostic exists for explicit edges, where an inconsistent loop means the configuration asserts contradictory amplifications and the posterior will split the difference somewhere the desk did not choose. The wire also carries, per node, the conditional $q_i$, the marginal precision, the `no_lit_path` mask and the anchor provenance — the three-quantity distinction of Exercise 3 made permanently visible.

# 8 Production orchestration

## 8.1 Modes, and what feeds the solve

`propagationMode` selects `smooth_field` (the current default — byte-identical to the pre-revamp engine, locked by the full legacy suite), `precision_messages`, or `hybrid` (message factors plus an explicit $\eta$-weighted legacy smoothness term; config-only, and locked to coincide with pure message mode at $\eta = 0$). The message path reuses everything the legacy path computes — selected universe, transported priors and provenance tiers, lit innovations and calibration precisions — and feeds the same `HandleField` seam back to reconstruction, functional bands, the idio floor, attribution and the backtest, which run unchanged. Auto relations over the selected universe: one calendar factor per adjacent expiry pair per ticker (canonical short receiver, (5) precision) and one beta-one cross factor per same-expiry ticker pair (constant precision, orientation-neutral at beta one). Persisted edge rules override auto; request edges override both. The observation-selection plan ports to the information form through the $\Sigma^+$ columns — the same rank-one algebra, no covariance-form detour.

## 8.2 Schema v2 and the direction migration

The edge schema was rebuilt rather than overloaded: `source/informer` → `target/receiver` naming, explicit `messagePrecision`, three handle betas, relation class, and a precision rule (`explicit` or `calendar_distance`, the latter re-deriving from (5) on every solve). The migration confronted a recorded trap: the legacy engine's row convention is $W$[from][to] with *"to" informing "from"* — the opposite of what parts of the schema documentation implied, hidden for years by symmetric topologies. The one-shot legacy import therefore maps new receiver = old `from` and new informer = old `to` — the labels invert so the economics do not (test-locked) — and is explicit, never silent: legacy weight is a relative trust with no canonical precision meaning, so the conversion factor is the caller's stated choice and the legacy blob round-trips untouched. The full edge editor ships with the revamp: per-relation precision and betas, relation class, inherited-versus-explicit display, implied-reverse chips, seed-from-auto, receiver diagnostics, and a deterministic scenario preview locked to the golden numbers of this note — so a desk can configure the three canonical examples and see these exact means before saving.

### 8.3 The six-node case file, revisited under message semantics

The original note's miniature universe — an SPX calendar chain, the sector ETF XLK, two technology names — reruns under the new operator with the empirical seeds and `learned` amplitude presets (fig. 7; lit: SPX 3M +1.00 vol point, XLK 3M +0.55). The table is generated by the production solve; the `desk` column shows the same universe at $\rho = 1$:

| Node | Role | $\widehat{z}$ (pts) | sd (pts) | desk $\widehat{z}$ | $q_i$ |
|---|---|---|---|---|---|
| `SPX1M` | dark | +0.66 | 1.37 | +2.97 | 1233 |
| `SPX3M` | lit | +0.95 | 0.10 | +0.99 | 51740 |
| `SPX6M` | dark | +0.11 | 0.71 | +0.49 | 4626 |
| `XLK3M` | lit | +0.54 | 0.10 | +0.56 | 31170 |
| `AAPL3M` | dark | +0.41 | 0.49 | +0.82 | 22166 |
| `MSFT3M` | dark | +0.41 | 0.49 | +0.82 | 22166 |

Every design choice is legible in the numbers. SPX 1M inherits the index move through the ladder at the $\alpha_T$=1 shape — +2.97 points under desk force, shrunk to +0.66 by the learned calendar level, with the widest band in the universe (short-dated vol units are where calendar noise is largest). AAPL and MSFT each hear two corroborating sources — index and sector ETF — so their transfer (+0.41; desk +0.82) sits *above* what either single-source level would give, exactly the corroboration lift of proposition 2. And the conditional $q_i$ column differs from the implied marginal everywhere — the Exercise-3 distinction on display.

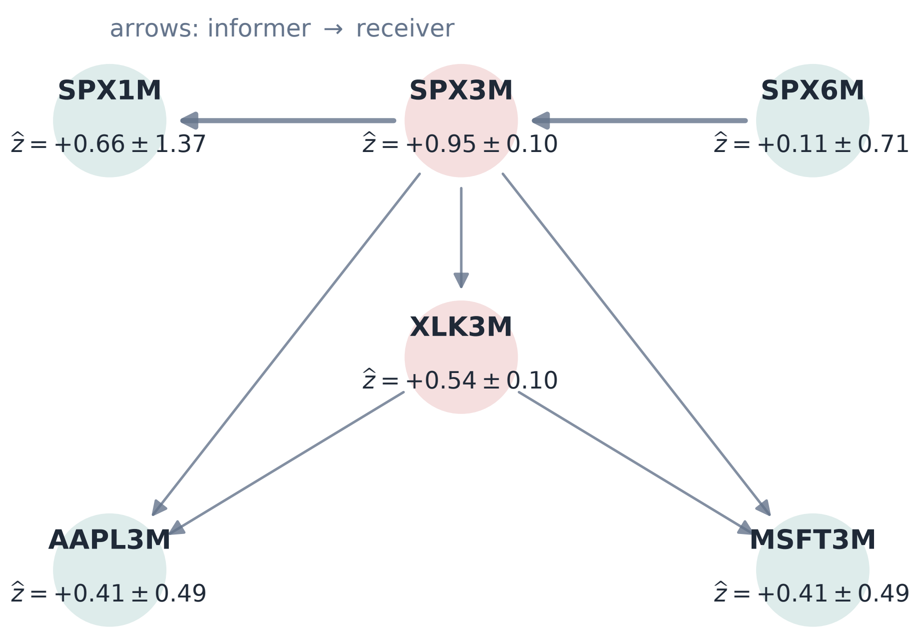


Figure 7: The six-node case file under message semantics (production solve; learned amplitude presets, Phase-0 precision seeds; arrows point informer → receiver). Lit nodes red, dark teal, each labelled with its posterior innovation and marginal sd in vol points. The names hear index and ETF corroboration and transfer above the single-source level; the 1M node shows the amplified-but-uncertain short end.

# 9 Evidence, recorded and in flight

## 9.1 What the legacy engine already proved

The smooth field's leave-one-out record is the floor any successor must clear, and it stands: on the 25-asset, three-regime benchmark ($\sim$47k held-out scores, transport-regime bracketed), fully-dark single names behind lit indexes and ETFs gain +7.9 to +14.2 ATM vol bps in the August-2024 spike and +3.8 to +7.2 fully out-of-sample in the October-2022 bear, with calm-regime skill +0.7 to +0.8 — never negative — and the once-dishonest calm dark-name bands closed by the idiosyncratic ATM floor (std $\zeta$ $1.91 \to 1.02$ and $1.85 \to 1.03$, mean-invariant by construction). Two recorded verdicts frame the revamp honestly. *The transient-name topology case file*: the harness once emitted cross-asset edges one-way, making single names transient states of the directed walk — $\pi_i = 0$, conductance zero, names disconnected, and a measured "0.01 bp response to a 96 bp index move" that was pure artifact until the reverse-edge fix; the new operator's $q_i$ and `no_lit_path` diagnostics would have named that failure on the first run, which is itself an argument for contractual semantics. *The item-14 closure*: the learned-beta sweep met its liquid-split activation rule but the gain was fractions of a bp (recorded, not wired), and the OT term at $\lambda = 1$ failed outright (negative skill, $\zeta$ std blown to 1.3–2.6) — the smooth field's levers were near their ceiling, which is why the redesign changes the operator rather than the knobs.

## 9.2 The Phase-0 design study and the smoke pair

The redesign's own empirical base is the stored-row study already cited throughout: transfer slopes 0.391 (index) and 0.762 (peer), the corroboration adjudication of fig. 2, the calendar seeds of fig. 3, and the golden fixtures — twenty-four contracts locked against a brute-force Gaussian reference *before* the operator existed, then reproduced through it at $10^{-12}$. One single-pair smoke of the full production path is recorded with the explicit caveat that one pair is not a verdict: message-learned full-LOO ATM skill $+5.8/+1.9$ bp against the legacy $+0.7/+0.3$, with the message bands slightly narrow ($\zeta$ std 1.2–1.4) where the legacy's ran wide (0.6).

## 9.3 The pre-registered gate

Precision messages become the product default *only if*, on the frozen three-regime fixtures over the strict out-of-sample window, the combined campaign shows: material liquid-split dark-name skill over both the transported prior and the current smooth field; no degradation in the stressed regimes; no calm-regime harm beyond tolerance; $\zeta$ std near one with 80/95% band coverage near nominal after the idio floor; no unstable cycles; and no wing-RMS deterioration. Expectations were recorded before the sweep: the `desk` preset is *expected* to lose day-horizon RMS (it is a belief mode and ships opt-in regardless); the `learned` preset is the candidate; a constant-precision ablation probes whether the gap-flat noise makes the decay term superfluous. The campaign — six variants absorbing the parked learned-betas sweep, band coverage added to the harness for the occasion — is running in the user's window as this note is built; its parts land beside the frozen fixtures, and the decision table will be filled from the artifact, not from enthusiasm. Until then the default stays `smooth_field`, and everything in this note is reachable one explicit mode switch away.

## 10 What is genuinely original here

Gaussian graphical models, information filters and gauge arguments are classical; the synthesis is specific. *Contractual propagation for volatility*: every desk-facing behaviour — full-amplitude transfer, precision addition, averaging, distance costing confidence not amplitude — is a golden test locked before implementation, so the operator's semantics cannot drift. *Amplitude as shape times measured level*, with the level mechanized through a node-linked anchor whose corroboration law was validated on stored rows to 0.3% with zero free parameters — a design decision adjudicated by a falsifiable prediction, not a preference. *The pairwise-factor rejection of the row form* on dead-informer grounds (fig. 5A): a joint-distribution defect invisible to every local conditional, caught and documented before it shipped. *Honesty rules as first-class machinery*: no-lit-path components left improper on purpose, baseline uncertainty placed exactly once, attribution by source never by path. And *evidence discipline*: the successor to a measurably skilful engine arrives with a pre-registered gate and an in-flight adjudication rather than a default flip.

## 11 Limitations

Where the guarantees stop. *The adjudication is not in*: every mean-semantics contract here is exact, but whether the message operator *predicts better* than the smooth field is precisely what the running campaign decides, and this note refuses to pre-empt it — today's default remains the smooth field. *Correlated informers overstate conditional precision*: $q_i$ adds configured precisions as if sources were conditionally independent; the global solve prices shared *routes* correctly, but shared *factors* (SPY and QQQ moving on one macro shock) need class discounts or a common-factor node — open, with the marginal-vs-conditional gap on the wire as the tell. *Cross-class seeds are upper bounds* (ticker-median measurement); per-edge noise will be higher. *The amplitude shape is weakly identified at the day horizon — $\alpha_T = 1$ is held by semantics, not by $R^2$. Hybrid mode is machinery, not a recommendation* (validated only at its $\eta = 0$ identity with pure messages). *The solve is dense*, sized for the $O(10^2$–$10^3)$-node selected universe; the factor list is sparse-ready and the sparse pass is deferred until a universe demands it. *Calendar arbitrage remains soft*: calendar factors propagate maturity signals but impose no hard cross-expiry projection — publish-time diagnostics and the LV projection remain the fence. And *curvature stays the weakest handle* end to end, in identification and in units, exactly as in the legacy engine.

## A Hyperparameter atlas

The only home for settings names: the body speaks mathematics, this table speaks configuration.

Table 2: Message-mode hyperparameters (legacy smooth-field knobs are unchanged and live in the original note's atlas).

| Knob | Default | Role |
|---|---|---|
| `graphPropagationMode` | `smooth_field` | The mode fork: `smooth_field` (byte-identical legacy) / `precision_messages` / `hybrid` (config-only). |
| `calendarBetaExponent` | 1.0 (per handle) | The amplitude shape $\alpha_T$ of (3). |

| Knob | Default | Role |
|---|---|---|
| `calendarAmplitude` | 1.0 | Calendar level $\rho$; presets desk = 1.0 / learned $\approx 0.23$ (the $\alpha_T$=1 value). |
| `crossAmplitude` | 1.0 | Shared level $\rho$ for every cross relation class in v1; learned $\approx 0.39$ index. |
| `calendarPrecisionScale` $p_0$ | $1.7 \times 10^3$ | Calendar precision scale (5), Phase-0 seed. |
| `calendarPrecisionEpsilon` $\epsilon_T$ | 0.97 | Near-identical- expiry precision cap; dominates at the day horizon. |
| `calendarPrecisionDecay` | `inverse_sqrt_gap` | Family: inverse-sqrt-gap / constant / log-distance. |
| `crossPrecisionScale` | $1.3 \times 10^4$ | Constant cross-relation precision (Phase-0 index seed; upper bound). |
| `innovationAnchorPrecision` | derived | Override for (4); unset = node-linked from $\rho$, and $\rho = 1$ gives $\kappa = 0$ exactly. |
| `cycleBetaTolerance` | $10^{-9}$ | Gauge-sweep flag threshold (proposition 3). |
| `messageEdges` / persisted rules | — | Request edges → persisted schema-v2 rules → auto relations, in that precedence. |
| *Hidden (module constants)* | | |
| `HANDLE_PRECISION_SCALE` | $(1, 0.36, 0.0036)$ | The $(s_\sigma/s_h)^2$ per-handle precision units of section 5. |
| `RELATION_CLASSES` | 5 classes | calendar / broad_index / sector_etf / sector_peer / custom. |
| `DISCONNECTED_Z_SD` | $(0.03, 0.05, 0.5)$ | The honest no-lit-path innovation sd per handle (one typical move). |
| canonical orientation | shorter $T$ | Auto calendar receiver; reverse identity $p_{\text{rev}} = p_{\text{fwd}}/\beta^2$. |

# B Performance notes

1. **The assembly is the factor list**: $O(E)$ triplets, dense materialization at the current $O(10^2$–$10^3)$-node design point, and the existing 1k-node perf rail passes unchanged. Sparse Cholesky / selected-inverse marginals are the deferred scale pass — the pairwise form was chosen partly because it sparsifies without renormalization coupling.

2. **Per-component solves**: the posterior inverts each factor- support component separately, so one disconnected block never pays for another, and the Cholesky PD guard localizes any conditioning failure to a named component.

3. **The message path adds no new data work**: universe, transported priors, innovations and calibration precisions are the same objects the legacy path computes; the mode fork is downstream of all of them.

4. **Adjudication runs where long jobs survive**: the campaign executes from the user's own shell (chunked, resumable parts beside the frozen fixtures); tool-spawned background jobs are killed on this box, which is a recorded operational constraint, not a suggestion.

5. No benchmark in this note was re-timed; figures and macros were generated at commit `aac01f4` on 2026-07-19, golden numbers through the production modules and empirical numbers from

the stored Phase-0 artifact.

## C Traceability

Table 3: Claims in this note and the code/tests that lock them.

| Claim | Object | Code anchor | Test anchor |
|---|---|---|---|
| Pairwise operator PSD; receiver conditional; $q_i$ unit mapping; canonical orientation | (2), proposition 1 | `volfit/graph/message.py` | `test_graph_message.py` |
| Golden contracts locked before code (transmission, competition, cross-asset, multi-hop, dead informer, repeated path, shrunk mode, baseline-once) | sections 3, 4 and 6 | `tests/fixtures/graph_message_golden.json` | `test_graph_message_golden.py` (brute-force reference), `test_graph_message.py` |
| Information-form component solve; no-lit honesty; reachability guard; exact attribution | (7) | `volfit/graph/message_posterior.py` | `test_graph_message_posterior.py` |
| Node-linked anchor $\kappa = p(1-\rho)/\rho$ fixed at build; $\rho = 1 \Rightarrow \kappa = 0$; corroboration law | (4), proposition 2 | `volfit/graph/message.py` | `test_graph_message.py`, `test_graph_message_golden.py` |
| Cycle gauge sweep flags inconsistent products | proposition 3 | `volfit/graph/message.py` | `test_graph_message.py` |
| Production orchestration: auto relations, $r^d$ harmonic, band placement, mode fork, hybrid $\equiv$ message at $\eta = 0$ | section 8 | `volfit/api/graph_message.py`, `api/graph_extrapolation.py` | `test_graph_message_production.py` |
| Schema v2; legacy-import direction inversion preserves economics; legacy blob untouched | section 8 | `api/schemas.py` (`GraphMessageEdge`), `api/graph_message.py` | `test_graph_message_production.py` |
| Smooth-field byte identity at defaults | invariant 6 | `volfit/graph/\{prior, posterior\}.py` | legacy suite unmodified (1302 green), `test_graph_example.py` |
| Adjudication harness variants, coverage metrics, decision table | section 9.3 | `backtest/run_message_adjudication.ps1`, `backtest/graph_edges.py` | `test_message_adjudication.py` |
| Legacy 25-asset verdict; transient-name topology fix; idio floor | section 9 | `backtest/graph_loo.py`, `volfit/graph/idio.py` | `test_graph_loo_backtest.py`, `test_graph_idio.py` |
| Phase-0 study (anchor choice, seeds, learned levels) | sections 4 and 5 | `backtest/message_phase0.py` | artifact `results/message_phase0.json` (regenerable) |

## D Reference implementation

The operator and the posterior are each a few lines. Executed against the production modules on a four-node directed universe with a node-linked anchor (operator matrix, posterior mean, marginal variances), the maximum deviation is $2.3 \times 10^{-13}$; the gauge sweep cross-check flags nothing on a consistent triangle (0 flags) and reports the planted inconsistent product 0.833 on the broken one.

Listing 1: The pairwise operator and the information-form solve (verified against `graph/message.py` and `graph/message_posterior.py`; agreement stated above).

```
def message_operator(n, factors):         # factors: (i, j, p, beta)
    Q = np.zeros((n, n))
    for i, j, p, b in factors:
        u = np.zeros(n); u[i] = 1.0; u[j] = -b
        Q += p * np.outer(u, u)           # PSD for any real beta
    return Q

def message_posterior(Q, kappa, obs, d, r):
    Qp = Q + np.diag(kappa)             # anchors (0 in desk mode)
    b = np.zeros(len(Q))
    Qp[obs, obs] += r                   # H^T R H
    b[obs] += r * d                     # H^T R d
    Sigma = np.linalg.inv(Qp)           # per lit component
    return Sigma @ b, Sigma             # z_hat, covariance
```

## References


[1] C. Rasmussen and C. Williams. *Gaussian Processes for Machine Learning*. MIT Press, 2006.

[2] D. Koller and N. Friedman. *Probabilistic Graphical Models*. MIT Press, 2009.

[3] J. Pearl. *Probabilistic Reasoning in Intelligent Systems*. Morgan Kaufmann, 1988.

[4] G. Peyré and M. Cuturi. Computational optimal transport. *Found. Trends Mach. Learn.*, 11(5–6):355–607, 2019.